80 mm (3.15 in) bore × 50 mm (1.97 in) rod Cylinder Force Data

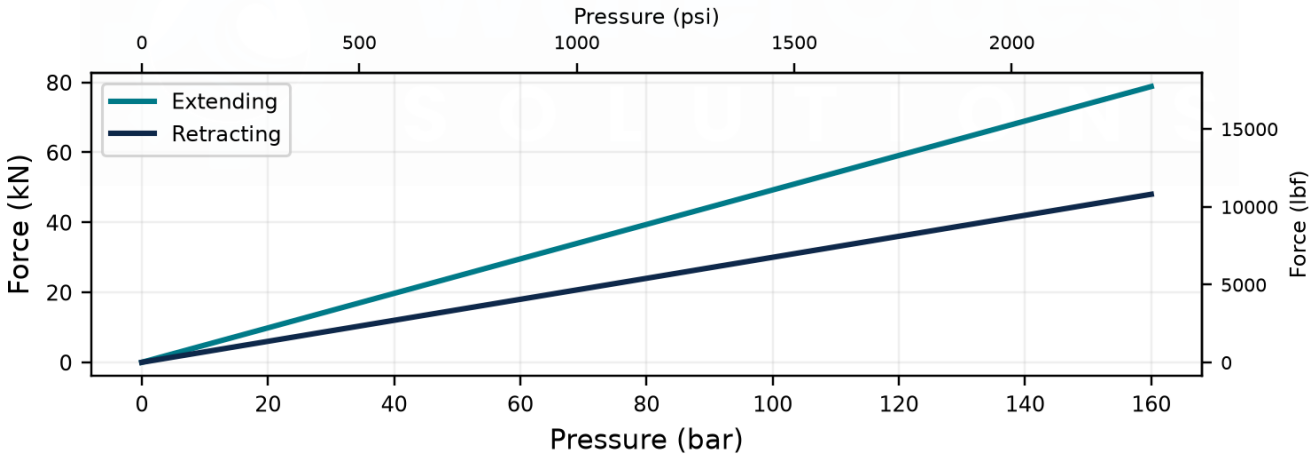
This geometry has a retraction area equal to 60.9% of its full piston area. Compare push and pull force at the same pressure, then check stroke, buckling, mounting and return-side pressure for the actual load.

Complete force output data across all operating pressures from 0 to 160 bar (0 to 2,321 psi). All values calculated with 0.98 efficiency factor.

At a glance

Parameter / value	Parameter / value
Piston Area 50.27 cm² (7.792 in²)	Ring Area 30.63 cm² (4.748 in²)
Max Extending Force 78.82 kN (17,719 lbf)	Max Retracting Force 48.03 kN (10,798 lbf)

Push and pull force



Plotted from the webpage force table, including its stated 0.98 efficiency factor.

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Pressure	Force Extending (kN)	Force Retracting (kN)
0 bar (0 psi)	0.00 kN (0 lbf)	0.00 kN (0 lbf)
10 bar (145 psi)	4.93 kN (1,108 lbf)	3.00 kN (674 lbf)
20 bar (290 psi)	9.85 kN (2,214 lbf)	6.00 kN (1,349 lbf)
30 bar (435 psi)	14.78 kN (3,323 lbf)	9.01 kN (2,026 lbf)
40 bar (580 psi)	19.71 kN (4,431 lbf)	12.01 kN (2,700 lbf)
50 bar (725 psi)	24.63 kN (5,537 lbf)	15.01 kN (3,374 lbf)
60 bar (870 psi)	29.56 kN (6,645 lbf)	18.01 kN (4,049 lbf)
70 bar (1,015 psi)	34.49 kN (7,754 lbf)	21.01 kN (4,723 lbf)
80 bar (1,160 psi)	39.41 kN (8,860 lbf)	24.01 kN (5,398 lbf)
90 bar (1,305 psi)	44.34 kN (9,968 lbf)	27.02 kN (6,074 lbf)
100 bar (1,450 psi)	49.26 kN (11,074 lbf)	30.02 kN (6,749 lbf)
110 bar (1,595 psi)	54.19 kN (12,182 lbf)	33.02 kN (7,423 lbf)
120 bar (1,740 psi)	59.12 kN (13,291 lbf)	36.02 kN (8,098 lbf)
130 bar (1,885 psi)	64.04 kN (14,397 lbf)	39.02 kN (8,772 lbf)
140 bar (2,031 psi)	68.97 kN (15,505 lbf)	42.02 kN (9,446 lbf)
150 bar (2,176 psi)	73.90 kN (16,613 lbf)	45.03 kN (10,123 lbf)
160 bar (2,321 psi)	78.82 kN (17,719 lbf)	48.03 kN (10,798 lbf)